

1.3 Classwork: Mean, Median and Mode — Markscheme

Data sources: Tax Foundation, VAT Rates in Europe 2026 and State and Local Sales Tax Rates 2026 (state rate only). Both sets are real 2026 figures.

1. [Maximum mark: 8]

Twelve standard VAT rates, sorted:

18, 19, 19, 20, 20, 21, 21, 21, 22, 23, 25, 27

$$n = 12, \sum x = 256.$$

(a) $\bar{x} = \frac{256}{12} = \frac{64}{3} = 21.333\dots = 21.3$ (3 s.f.)

M1A1

M1 for a sum of the twelve values divided by 12, by hand or on the GDC; A1 for 21.3. Accept $\frac{64}{3}$ or $21\frac{1}{3}$. Do not accept 21 — that is the median and the mode.

GDC: menu → **Statistics** → **Stat Calculations** → **One-Variable Statistics**.

(b) With $n = 12$ the median is the mean of the 6th and 7th values: $\frac{21 + 21}{2} = 21$

M1A1

M1 for an ordered list or for identifying the middle pair; A1 for 21. A candidate who takes a single middle value still reaches 21 here — award both marks, but the return should say that with an even n the two middle values are averaged.

(c) Mode = 21 (Spain, Netherlands and Belgium).

A1

Do not accept 3, which is the frequency of the mode.

(d) The distribution is **skewed right** (positive skew).

A1

Because $\bar{x} = 21.3 > \text{median} = 21$, and the dot plot has a longer tail on the high side while most values bunch between 19 and 22.

R1

R1 for a justification of either kind — the mean exceeding the median, or the long tail on the right. Naming the shape with no reason earns the A1 only. “Skewed left” earns nothing, however it is justified. FT: award the R1 for a comparison that is correct for the candidate’s own mean and median.

(e) **Hungary** at 27%: it sits well above the rest, with a gap between it and the next value (25%), far from where the bulk of the data lies.

R1

R1 requires the country (or the value 27) and a reason of the right kind — distance from the rest of the data, or the gap. “It is the biggest” earns nothing: every data set has a biggest value. No $1.5 \times$ IQR test is expected here; the definition in use is informal.

2. [Maximum mark: 8]

Thirteen state sales tax rates, sorted:

0, 0, 0, 2.9, 4.0, 4.0, 4.0, 4.0, 5.3, 6.0, 6.25, 7.0, 7.25

$n = 13$, $\sum x = 50.7$.

(a) $\bar{x} = \frac{50.7}{13} = 3.9$ exactly

M1A1

M1 for a sum of the thirteen values divided by 13; A1 for 3.9. The commonest error is to drop the three zeros and divide by 10, giving 5.07: that earns the M1 only. A state charging 0% is still a data value.

(b) With $n = 13$ the median is the 7th value: 4.0

M1A1

M1 for an ordered list or for identifying the 7th position; A1 for 4.0.

(c) Mode = 4.0 (Alabama, Georgia, New York and Wyoming).

A1

Four states share 4.0; three share 0. Do not accept 0, and do not accept the two quoted together — the frequencies are not equal.

(d) The distribution is **skewed left** (negative skew).

A1

Because $\bar{x} = 3.9 < \text{median} = 4.0$, and the dot plot has a tail running down to the three states at 0 while most values sit between 4 and 7.25.

R1

The contrast with Problem 1 is the point of the pair: same three statistics, opposite order, opposite shape. FT as in 1(d).

(e) Either answer earns the mark if it is argued.

R1

Accept “outliers”, because they sit far below everything else and distance from the rest of the data is what makes a value an outlier. Accept “a separate group”, because there are three of them at exactly the same value and those states have chosen to have no sales tax at all, which is a policy rather than a stray measurement. Accept “both”. The mark is for the reasoning; a bare label earns nothing. This is the open interpretation item.

Teacher notes — not for the student sheet

Local taxes on top. The Problem 2 figures are state rates only. Cities and counties add their own: the population-weighted national average combined rate is about 7.5%, and Louisiana averages about 10.1% combined against a 5.0% state rate. Alaska appears at 0% here but its boroughs do levy local sales tax — a good example of a data value that is true as recorded and misleading as read.

How the thirteen states were chosen, and what that costs. The selection is pedagogical, not statistical — a convenience sample picked to produce a clean left tail, a single mode, an odd n and a sum that divides exactly. It is not representative, and the numbers say so:

	n	Mean	Median	Mode	% at 0
The thirteen on the sheet	13	3.90	4.00	4.0	23%
The other 37 states	37	5.51	6.00	6.0	5%
All 50 states	50	5.09	6.00	6.0	10%

The sheet's mean sits 1.2 points below the true 50-state mean and its median a full 2 points below. Three of the five zero-rate states are in (Alaska and New Hampshire are not), so zeros are over-represented more than twofold. The real mode is 6.0% — ten states charge it — and only one of those ten (Florida) is on the sheet. Eight of the thirteen rank in the bottom twenty of fifty; nothing sits between rank 5 and rank 15.

What survives. *The conclusion does. All fifty states are also skewed left, and more sharply than the sheet: mean — median is -0.91 across all fifty against -0.10 for the thirteen. So the shape students find is the real shape; only the statistics are shifted. That is the briefing, and it ties straight back to 1.2 — this data set is exactly the kind of purposive sample they were asked to criticise last week, and it still gets the qualitative answer right. Worth asking them why a biased sample can preserve a shape while moving every number.*

City income tax, for the discussion about putting two levels on one graph. *New York City residents pay a city income tax on top of New York State's, at four rates from 3.078% to 3.876% (single filers: 3.078% on the first \$12,000 of city taxable income, then 3.762%, 3.819% and 3.876% above \$50,000). Yonkers levies a surcharge as well. Most US cities levy none at all. Two ways to put that on a graph are worth raising with the class: stack the city rate on the state rate as a bar with two segments, so the total is the bar height and the split is visible; or plot state rates alone and mark the few cities separately, since a variable that is zero for almost every city is not really the same variable as the state rate. The first answers "what does a resident pay", the second answers "how do states differ" — which question you are asking decides the graph.*